

2-15 Use an Algorithm to Subtract

Learning goal: I can subtract multi-digit numbers with an algorithm by lining up the places and subtracting from the ones to the left, and check with addition.

Textbook: pages 91–94. The book subtracts $8,386 - 7,185$ one place at a time. Every problem in this lesson needs no regrouping.

Words to Know

difference	The answer when you subtract.
algorithm	Step-by-step column subtracting: start with the ones and move left.
vertical	Written in columns, one number under the other.
line up	Put ones under ones, tens under tens, and so on.
increase	How much a number grew. Find it by subtracting.

How to Teach It

1. Write the numbers vertically. Line up the **ones** first, even if one number is shorter.
2. Subtract the ones, then the tens, then the hundreds, then the thousands (top digit minus bottom digit).
3. If a place comes out 0, write the 0 so the other digits stay in their places (4,020).
4. If the bottom number has no digit in a place, write the top digit down (6,957 – 615: the 6 thousands stay).
5. Check: difference + the number you took away = the number you started with.

Tip: This lesson has no regrouping (every top digit is at least as big as the bottom digit). Subtraction with regrouping is practiced with the strategies in Lessons 2-7, 2-8, and 2-11; regrouping in the algorithm comes later, so there is no need to teach borrowing here.

Worked Examples

Example 1: Subtract place by place

$$\begin{array}{r} 9,486 \\ - 6,253 \\ \hline 3,233 \end{array}$$

Ones: $6 - 3 = 3$
 Tens: $8 - 5 = 3$
 Hundreds: $4 - 2 = 2$
 Thousands: $9 - 6 = 3$

Example 2: Line up the ones

$$\begin{array}{r} 5,867 \\ - 742 \\ \hline 5,125 \end{array}$$

742 has no thousands, so write the 5 thousands down.
 Do not line up on the left!

Example 3: Zeros and checking

$6,438 - 2,418 = 4,020$ ($8 - 8 = 0$ ones; $3 - 1 = 2$ tens; $4 - 4 = 0$ hundreds; $6 - 2 = 4$ thousands).

Check: $4,020 + 2,418 = 6,438$ ✓

Example 4: Missing digits

$$\begin{array}{r} 3,8\Box6 \\ - 1,\Box4\Box \\ \hline \Box,513 \end{array}$$

Ones: $6 - 3 = 3$. Tens: $5 - 4 = 1$.
 Hundreds: $8 - 3 = 5$.
 Thousands: $3 - 1 = 2$.
 $3,856 - 1,343 = 2,513$

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|--|---|
| ☐ Line up numbers by place, starting with the ones. | ☐ Check a difference with addition. |
| ☐ Subtract from right to left and write any 0s. | ☐ Find missing digits in a subtraction problem. |
| ☐ Subtract 3-, 4-, and 5-digit numbers. | ☐ Solve “how many left,” “other part,” and “which grew more” problems. |

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Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Line up the ones. Then use an algorithm to subtract $9,365 - 142$.

Difference: _____

- B.** A school play sold 7,968 tickets for two shows. It sold 4,725 tickets for the first show. How many tickets did it sell for the second show? Use an algorithm.

Tickets: _____

- C.** Find $5,439 - 2,108$. Then check your answer with addition.

Difference: _____

Check: _____

On My Own *Try these by yourself. Show your work.*

- 1.** Use an algorithm to subtract.

$$\begin{array}{r} 957 \\ - 314 \\ \hline \end{array}$$

- 2.** Use an algorithm to subtract.

$$\begin{array}{r} 685 \\ - 263 \\ \hline \end{array}$$

3. Use an algorithm to subtract.

$$\begin{array}{r} 4,879 \\ - \quad 456 \\ \hline \end{array}$$

4. Use an algorithm to subtract.

$$\begin{array}{r} 6,857 \\ - 3,807 \\ \hline \end{array}$$

5. Use an algorithm to subtract.

$$\begin{array}{r} 7,596 \\ - 4,284 \\ \hline \end{array}$$

6. Use an algorithm to subtract.

$$\begin{array}{r} 47,596 \\ - 3,284 \\ \hline \end{array}$$

7. A family is driving 3,785 miles. They have driven 562 miles. How many miles do they have left to drive?

8. A club spent \$5,896 on a trip and food. The trip cost \$2,473. How much did the club spend on food?

Miles: _____

Food: _____

9. Fill in the missing digits. Explain how you found each one.

$$\begin{array}{r} 6, \boxed{} 47 \\ - 3, 5 \boxed{} \boxed{} \\ \hline \boxed{}, 435 \end{array}$$

10. A school fundraiser raised this much money each year. Was the increase greater from Year 1 to Year 2, or from Year 2 to Year 3? Explain.

Year 1	Year 2	Year 3
\$1,423	\$3,645	\$5,856

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Together

A	9,223 <i>Line up 142 under 365: $5 - 2 = 3$, $6 - 4 = 2$, $3 - 1 = 2$, then write the 9.</i>
B	$7,968 - 4,725 = \mathbf{3,243}$ tickets
C	3,331 . Check: $3,331 + 2,108 = 5,439 \checkmark$

On My Own

1	643
2	422
3	4,423 <i>Watch the lining up: 456 goes under 879.</i>
4	3,050 <i>Write the zeros: ones $7 - 7 = 0$; tens $5 - 0 = 5$; hundreds $8 - 8 = 0$; thousands $6 - 3 = 3$.</i>
5	3,312
6	44,312 <i>The 4 ten-thousands come straight down.</i>
7	$3,785 - 562 = \mathbf{3,223}$ miles
8	$\$5,896 - \$2,473 = \mathbf{\$3,423}$
9	$6,947 - 3,512 = \mathbf{3,435}$ <i>Ones: $7 - 2 = 5$. Tens: $4 - 1 = 3$. Hundreds: $9 - 5 = 4$. Thousands: $6 - 3 = 3$. Check: $3,435 + 3,512 = 6,947$.</i>
10	Year 1 to Year 2: $3,645 - 1,423 = 2,222$. Year 2 to Year 3: $5,856 - 3,645 = 2,211$. Year 1 to Year 2 had the greater increase. $2,222 > 2,211$ (compare the tens: $2 > 1$).